

Software Requirements Specification

Domain & SSL Certificate Expiry Alert System

1. Document Control

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2. Purpose & Scope

This documents specifies the functional and non-functional requirments for an IT operations utility that performs automated daily WHOIS registration and SSL/TLS handshake, alerting system administrators through an escalation ladder before certificate expiration.

3. Actors

- System Admin – configures monitored domains, receives day to day expiry alerts, and reviews audit logs.
- Security Officer – receives escalated alerts when a System Admin fails to acknowledge a critical expiry warning.

4. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall initiate TLS handshakes against monitored domains daily, extract certificate expiration dates and trigger alerts at 3, 15 and 30 days before expiry.	High	Pass: Alert queued for certificate expiring in 15 days. Fail: Expired certificate ignored without notification	Core function of the monitoring system Prevents unnoticed certificate lapses.

FR-002	Functional	The system shall perform a daily WHOIS lookup on each monitored domain and extract the domain registration expiry date.	High	Pass: System retrieves and stores an accurate expiry date for a domain expiring within 90 days. Fail: Expiry date is missing or older than 24 hours.	Domain expiry is a distinct failure mode from certificate expiry and requires independent tracking.
FR-003	Functional	The system shall allow a System Admin to add, edit, or remove domains and endpoints from the monitored list through a configuration interface.	Medium	Pass: A newly added domain appears in the next scheduled scan cycle. Fail: Domain is added but not scanned within 24 hours.	The system has no monitoring targets without a way to manage the watchlist.
FR-004	Functional	The system shall escalate an unacknowledged expiry alert to the Security Officer if the System Admin has not acknowledged it within 24 hours of the 3-day threshold alert.	High	Pass: Escalation notification is sent exactly once, 24 hours after an unacknowledged 3-day alert. Fail: No escalation occurs after 24 hours of silence.	Implements the escalation ladder required by the problem statement; prevents a missed alert from becoming a missed renewal.
FR-005	Functional	The system shall log every scan, alert, and escalation event with a timestamp, and make the log viewable by the System Admin or Security Officer.	Medium	Pass: Querying the log for a domain returns all scan, alert, and escalation events in chronological order. Fail: An event occurred with no corresponding log entry.	Provides an audit trail for compliance review and troubleshooting missed renewals.
NFR-001	Nonfunctional - Performance & Security	The monitoring engine shall scan a list of 1,000 domains and SSL endpoints in under 3 minutes.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures the system scales to enterprise-sized domain portfolios without delaying alerts.
NFR-002	Nonfunctional - Reliability	The system shall retry a failed	Medium	Pass: A domain with 2 transient	Prevents transient network

		WHOIS or TLS handshake attempt up to 3 times with exponential backoff before marking the domain as unreachable and raising an alert.		failures is still scanned successfully on the 3rd attempt. Fail: A single transient failure incorrectly marks the domain unreachable.	issues from generating false unreachable alerts during the daily automated scan
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